Evidence Synthesis Methodology

Purpose:

An evidence synthesis (ES) is a branch of evidence-based-medicine (EBM) that allows the combining of information between qualitative and quantitative studies. used combining information from multiple studies that may have investigated similar research aim(s) leading to a comprehensive understanding to determine the prevalence or the effectiveness of a treatment. As such, an ES could include multiple synthesis and analysis methods such as a thematic, narrative and meta-analysis. The synthesis and analysis methods described in the context of this publication is led by a systematic methodology where a clear protocol comprised of an eligibility-criteria with MeSH terms and a suitable refinement method. In the context of this publication, the syntheses provide a rigorous and transparent knowledge base for translating any knowledge and practice gaps to develop novel research, policy briefs or clinical practice guidelines.

Methodology:

The use of ESPs are well established and such an example if the NIHR Health Technology Assessments that mainly uses quantitative and qualitative data from clinical trials to assess clinical effectiveness, health economic evaluations and efficiency of interventions to support healthcare decisions (NIHR et al., 2019). Therefore, ESPs are mixed-based methodologies to better identify, synthesis and report conclusions to support evidence-based-decisions (EBD). It is equally important to indicate, evidence synthesis are not conducted for the soul purpose of testing interventions (diagnostics or treatments) but to identify and report prevalence which is a vital step prior to systematic assessing medical interventions. For this reason, this paper uses a mixed-based evidence synthesis to evaluate the prevalence of Mental Health (MH) outcomes in among endometriosis women was developed. Furthermore, where possible, the identification of the possible disease sequalae among caucasian and BAME populations would be reported as part of the meta-analysis, if sufficient data is available.

Research Questions:

- Identify and report the prevalence on the sequalae of specific MH outcomes (symptomatologies and/or psychiatric comorbidities) and endometriosis
- Identify and report on the MH impact of endometriosis amongst BAME and non-BAME women

Hypothesis: Endometriosis populations may be more susceptible to a mental health impact, as such a possible disease sequalae should be explored. This evidence should be sufficient to further explore future research to better evaluate the mechanistic composition between endometriosis and mental health outcomes such as depression and/or anxiety with a view to use the aetiological understanding to better manage patients.

Eligibility/Data collection: A mixed-methods approach was used to complete the data analysis. A systematic protocol was created, peer reviewed and published on PROSPERO. This included MeSH terms and variables in line with the research questions such as depression, PTSD, anxiety, psychosis, endometriosis mood disorders, low mood, psychological distress, unipolar, bipolar and BAME. A snowball method was then applied to each of the MeSH terms to ensure a thorough data collection was completed. As per the eligibility criteria, all randomised controlled trials (RCTs) and non-RCTs peer reviewed and published in English. Multiple databases of Pubmed, Science direct, EMBASE, ProQuest and PROSPERO was used. The data extraction utilised Endnote and Microsoft Excel which was assessed independently for its integrity alignment to the research questions pre and post statistical analysis. A quality assessment was performed using a Newcastle-Ottawa-Scale (NOS).

Refinement protocol

Researchers that extracted the data and independent reviewers followed the steps outlined within the listed refinement steps below to gather, check and verify the retrieved data.

Step 1; All the data was administered within an excel spreadsheet once the dataset was ‘cleaned’ using Endnote

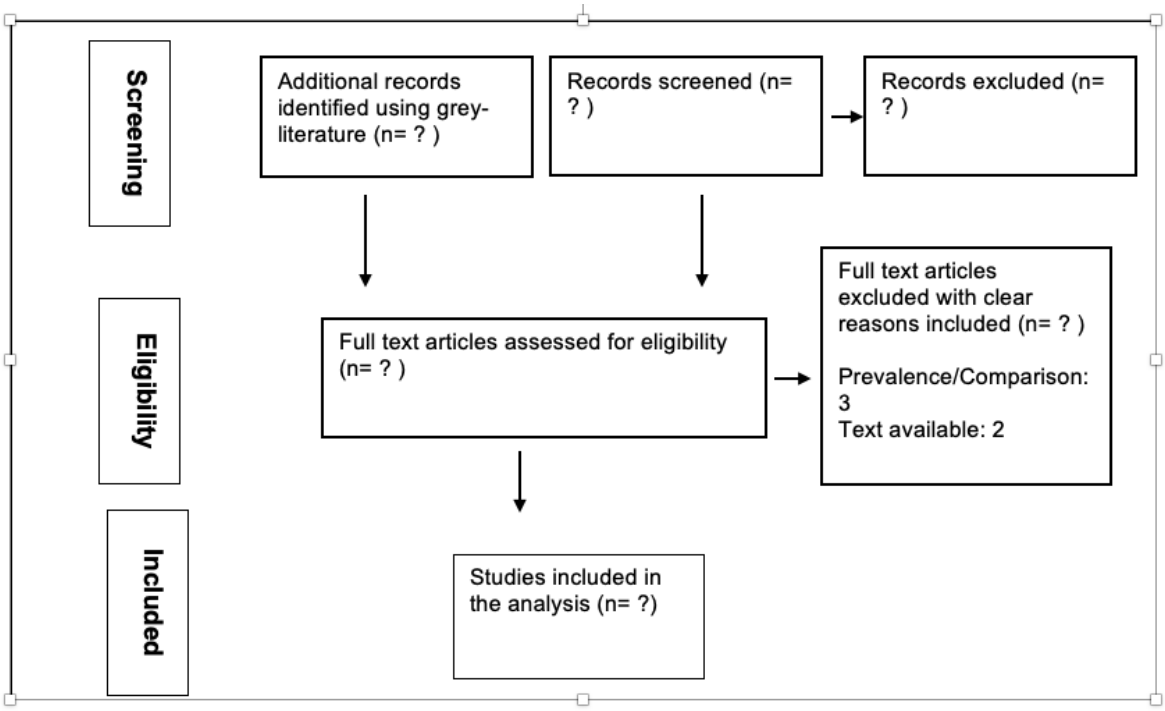
Step 2; Refinement step 2 was completed using the table below. The each study was reviewed for the variables of mental health outcomes reported either independently or in conjunction with pain disorders such as dyspareunia. Race and ethnicity was also gathered if needed.

Paper Title/initials of the first author/Year	Variables	Pre-existing condition Yes/No	Direct result of Covid Yes/No	Symptomology	Treatment received; Yes/No	Primary care or Secondary care; Yes/No
	Depression					
	Anxiety					
	Low mood					
	Psychosis					
	Dyspareunia					

Note; Each variable was reported as above where some studies may have multiple outcomes. Other additional data were recorded if these were in line with the research question.

Step 4: Completed tables were checked for any errors by the independent reviewers.

Step 5: All data were reported based on PRISMA



Data analysis:

Multiple methods were used as part of the data synthesis to ensure the data gathered is reported using the most appropriate analysis method;

- Thematic and narrative synthesis: common MH denominators were identified and reported using a systematic method, as per the protocol. Variables and key clinical features reported by the included studies were discussed within this section.
- Systematic review (SR) and meta-analysis (MA): Interpretation of the data was aligned to the relevance to the research questions. Eggers tests for publication bias were completed. NOS was used to assess the quality of the studies. The SR portion of the paper was reported as per the PRISMA tool. The SR protocol was peer reviewed and published in PROSPERO (CRD42020181495)
- In order to assess prevalence, an important composite would be to determine the true mental health impact and endometriosis that may or may not demonstrate mechanistic relationships. Upon review, if there is minimal evidence associated with a mechanistic relationship, the sequalae would be assessed by better evaluating disease burden. This could be considered between the BAME versus non-BAME populations although it is recognised

current studies do not always report race and/or ethnicity specific outcomes sufficient for a meta-analysis. In the event this information is minimal, a narrative synthesis would be completed.

- Most clinical researchers use PRISMA or PCIO as part of reporting guidelines although, this SR has been extension to other areas such as thematic and narrative composites to better discuss prevalence as a primary outcome. Therefore, PRIMSA has been used extended further.
- Heterogeneity would be assessed using I_2. Although, there are limitations especially if the datasets are large to determine prevalence (Rucker et al 2008), in the case of this SR/MA, it was relevant given the limited number of studies identified.

References

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